



Getting started with mqTrains Relay 1 Board

mqTrains OVERVIEW

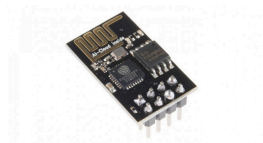
ESP8266 microprocessors are an inexpensive option for controlling model railroad accessories such as turnouts, sensors and signals. They connect quite easily to JMRI using the ESP's WiFi interface and the MQTT messaging protocol. ESP01s can be purchased from online stores for about a couple of bucks each.

mqTrains sketches have been developed or are in development to handle some common model train requirements such as turnout control, signalling and occupancy detection. Visit mqTrains.com for more information.

ESP devices only use the 2.4 GHz WiFi band. Some common domestic WiFi routers are limited to 16 or 32 concurrent sessions. Check your router specifications if you intend to have a high number of these devices.

mqTrains Relay 1 Board

The mqTrainsRelay1 sketch is designed for the popular single relay ESP01 carrier boards (usually sold as "ESP-01S Relay Module"). It drives the relay over MQTT and adds one input, so the same little board can also report a push button, toggle switch or occupancy detector back to JMRI.



- Relay output on GPIO0 – MQTT item /01
- Input on GPIO2 (active low) – MQTT item /02

LET'S GET STARTED

These are the main steps to get started with this mqTrains board. More complex features are covered in the [mqTrains User Reference](#).

1. Install the MQTT Server Software

Linux, Mac and Pi users:

We recommend using a Raspberry Pi for the MQTT Server Software. The Pi Zero W is quite adequate and inexpensive. Follow the vendor instructions to set up the Raspberry Pi. To install Mosquitto Server, use the commands:

```
sudo apt-get update
sudo apt-get install mosquitto mosquitto-clients
```

Mosquitto version 2 (since December 2020) uses tighter security rules. Create this config file:

```
sudo nano /etc/mosquitto/conf.d/mosquitto.conf
```

Insert these two lines and then press ctrl-o <enter> and ctrl-x

```
listener 1883
allow_anonymous true
```

and restart mosquitto with

```
sudo service mosquitto restart
```

Windows users:

Mosquitto can also be installed on most Windows computers. Refer to the [Mosquitto website](#) for details. If you install on Windows, you will most likely need to grant Firewall permission. Instructions for granting permission with the Windows Defender Firewall are shown at the end of this document. If you have a Firewall with non-Microsoft security software, refer to the vendor instructions.

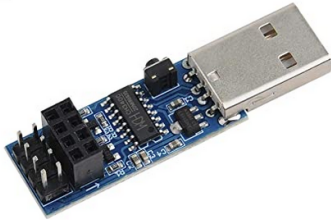
2. Install mqTrains on your ESP01

Download the mqTrains binary file from here: mqTrains.com/downloads/

Windows users: Download the [nodemcu-flasher](#) program from the appropriate link: x64 version (most common) or x32 version (for old 32-bit computers).

The ESP01 requires an adapter to connect to a computer. We recommend the one shown here. Point

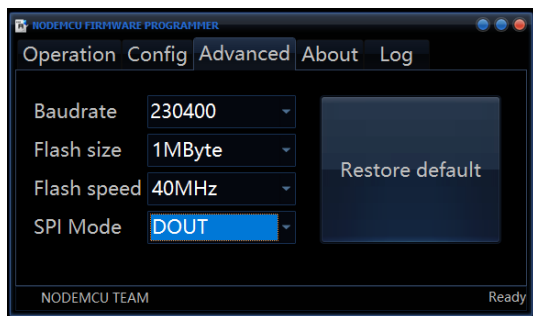
the ESP01 towards the USB connector.



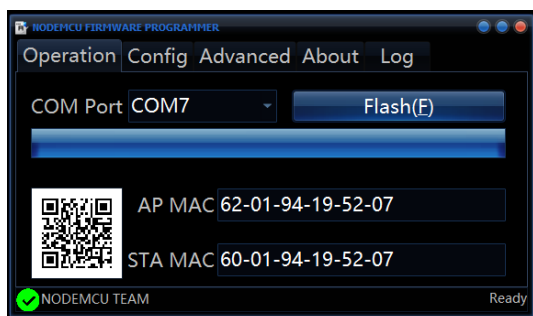
Run the installer (double click the exe file downloaded), select the Config tab, click on the wheel in the first row, then navigate to and select the binary file you downloaded.



Select the Advanced tab and select DOUT for SPI Mode:



On the Operation tab, the correct COM Port will most likely be selected for you, change if need be, then click on Flash to start the upload. The blue bar will show the upload progress which takes about 3 minutes. Wait for the green tick and "Ready" message at the bottom of the screen.



Linux, Mac and Pi users:

```
sudo apt-get update
sudo apt-get install python python-pip
sudo pip install esptool
```

Then, while in the folder with the .bin file, run (all in one line and with the correct serial port, ttyUSB0 shown here):

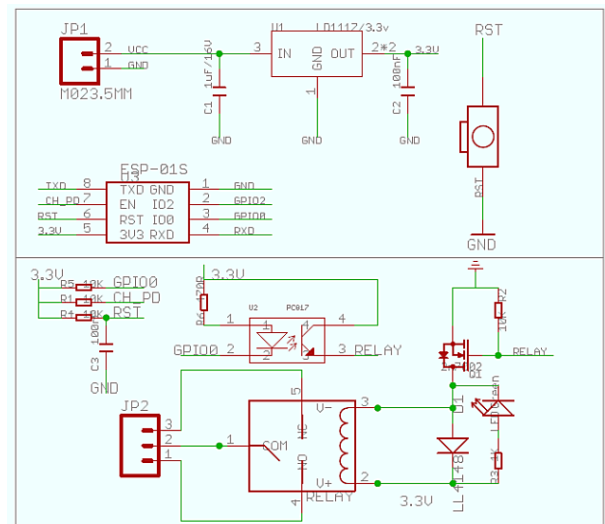
```
esptool.py --chip esp8266 --port /dev/ttyUSB0
--baud 460800 --before default_reset --after h
ard_reset write_flash 0x0 mqTrains_Relay1_x.xx
_1MB.bin
```

3. Wiring your ESP01

Plug the ESP01 into the relay module socket and connect a 5 Vdc power supply to the module's 5V and GND terminals. The module's on-board regulator powers the ESP01.

The relay contacts (COM and NO, or COM and NC) switch your accessory: a turnout motor, lighting circuit, or any load within the relay rating.

For the input, wire a push button, toggle switch or occupancy detector output between the ESP01's GPIO2 pin and Ground. The input is read as Active when pulled low (this can be inverted on the Relay Module Setup page).



Typical ESP01 relay module schematic

4. Configuring your ESP01

Using a smartphone or computer, look in your available WiFi networks list for the mqTrains Access Point (AP) network service started by the ESP. The default SSID name is "mqTrains-5001" with the Key or password set to "mqtrains".

Once connected to the ESP's AP, use a web browser to connect to the mqTrains configuration page using <http://2.2.2.1>.

You can use these QR codes to connect to the default Access Point and web page. (The 2nd QR code for the web page won't work if your app needs the internet to decode, like Google Lens.)



WiFi mqTrains-5001



Web Page 2.2.2.1

Select the Quick Start option from the main menu and change each of these values to suit your needs.

mqTrains Relay Board #5001

Quick Start

Board Serial Number:
(4 characters)
5001

MQTT Server IP address:
10.10.10.13

MQTT Server IP Port:
1883

Base Topic:
myLayout/turnout/5001

WiFi SSID:
TsNamib

WiFi Key (8+ chars):

SAVE+REBOOT MENU

0.12_Relay1wInput, -52

Set a 4 digit Serial Number for the board. If you have more than one mqTrains board, each should have a unique serial number. This will be used as part of the MQTT messaging. We recommend not leaving the default value, since the next new board will also be 5001. Some use 1000s for turnout boards and 2000s for sensor boards, others use two numbers for a location on the layout and two more for a serial number at that place, so why not change it to 5001?

Set the IP address of the computer on which your MQTT Server (Broker) is installed and the port number it uses. 1883 is the default port.

Set the base portion of the MQTT topic. The full topic will have the board serial number and the two digit item number appended to the base. For example, using the

default base topic, the full MQTT topic for the relay on board number 5001 will be myLayout/turnout/5001/01 and the input will report on myLayout/turnout/5001/02.

Set the WiFi SSID and Key to connect to your desired WiFi service. Keep in mind that the WiFi service will need to provide you access to the MQTT Server whose IP address you entered earlier.

Press SAVE+REBOOT to save all these values to the ESP's internal storage and restart it using those values.

Using a utility such as MQTT Explorer or an MQTT subscribe line command, look for messages for topics beginning with the Base Topic. Expect to see a message like "192.168.1.10 is here!" This is the IP Address of the ESP on your WiFi network. Use this IP address in your browser to connect to the configuration page to complete the set up or to make changes. The AP will no longer be needed and will shut down once the ESP is able to post a message to the MQTT Server.

Prefer a serial cable? These boards can also be provisioned over the serial port (115200 baud): send :h; for the command list, then for example :Bserial 9001; ;Bbase myLayout/relay/; and :Bsskr0000 MySSID MyWiFiKey; followed by a reboot.

5. Configure the relay and input

Select *Relay Module Setup* from the main menu. Tick *Enabled* for the relay and/or the input as required, and *Locked* if a channel must ignore MQTT commands for a while.

mqTrains Relay Board #5001

Relay Module Setup

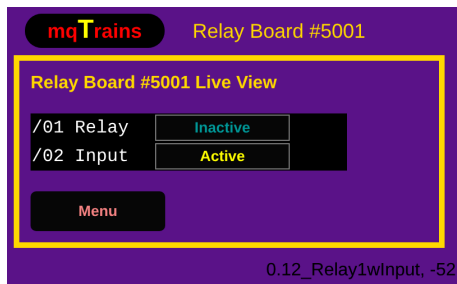
Relay is GPIO0 on topic /01, Input is GPIO02 on topic /02

	State	Enabled	Locked
/01 Relay	Inactive	☒	☐
/02 Input	Active	☒	☐

SAVE Live MENU

0.12_Relay1wInput, -52

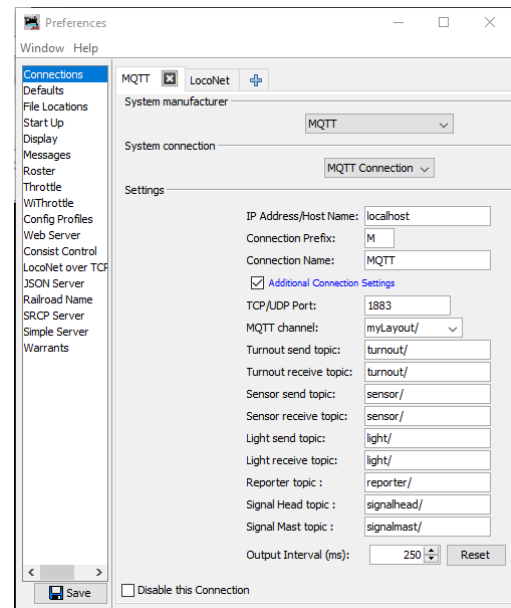
The states can be monitored using the Live View page. You can also toggle the relay by tapping on its state box. This publishes an MQTT message so JMRI can see that the state has changed and update accordingly.



The relay listens for the payloads THROWN, ON, ACTIVE or 1 to energise, and CLOSED, OFF, INACTIVE or 0 to release. TOGGLE (or 2) flips the current state. The words the board itself publishes (default ACTIVE/INACTIVE) can be changed on the MQTT Info page to THROWN/CLOSED if you prefer pure turnout messaging.

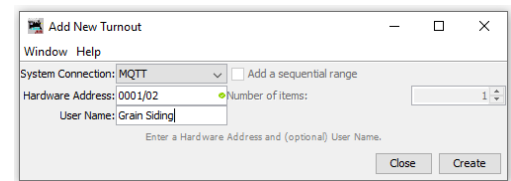
6. Set JMRI to talk with your ESP01

In JMRI, using Edit -> Preferences -> Connections, create an MQTT connection with the settings as shown, and especially make sure you specify the same host on which you have installed the MQTT Server as you provided to the ESP. You can specify the host name, or the IP address as you did on the ESP01. If the MQTT server is running on the same computer as JMRI, you can also use localhost as shown.



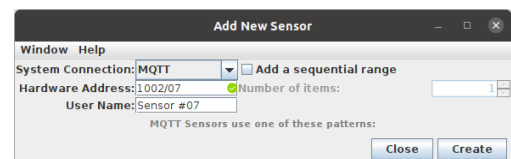
JMRI Connections (v 4.21.2)

Create an MQTT turnout for the relay using the board serial number and the 2 digit item number as hardware address, for example 5001/01.



JMRI Add New Turnout

Create an MQTT sensor for the input with hardware address 5001/02. When the input pin changes to Active, the sensor in JMRI will go ACTIVE.



JMRI Add New Sensor

Try mqTrains from the command line

The mqTrains web pages allow you to see and change the relay state. On a command line, with the mosquitto tools installed, you can see the messages being published while mqTrains is connected to the MQTT Server:

```
prompt> mosquitto_sub -v -h localhost -t "myLayout/#"
or in Windows
"c:\Program Files\mosquitto\mosquitto_sub.exe" -v -h localhost -t "myLayout/#"

-v means verbose - show the full message (topic and payload).
-t means topic
-h means the MQTT Server hostname or IP address
-r means retain the message in the Server.
-m means message, aka payload.
```

You can publish messages to command the relay and see the state change on the web page or in JMRI:

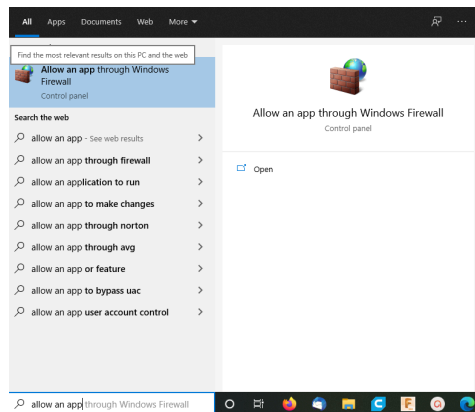
```
prompt> mosquitto_pub -h localhost -r -t "myLayout/turnout/5001/01" -m "THROWN"
prompt> mosquitto_pub -h localhost -r -t "myLayout/turnout/5001/01" -m "CLOSED"
```

Note that xxxx/00 is used for board specific things (see the mqTrains-UserReference.pdf for more information).

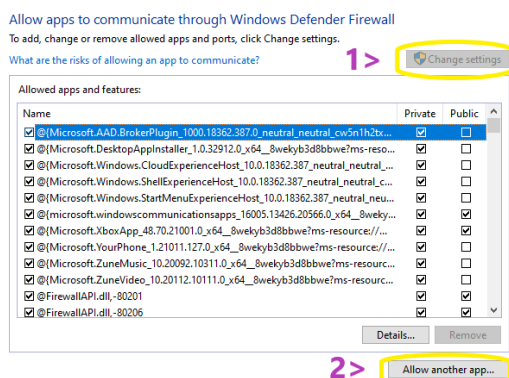
Burning through the Windows Firewall

If you install MQTT Server Software (mosquitto) on a Windows computer, such as your JMRI machine, you may find that the ESP nodes cannot connect to the MQTT Server due to the Windows Defender Firewall blocking WiFi traffic coming into the computer. This section describes how to alter the Firewall to allow MQTT messages to pass through.

Type "allow an app" into the Windows search bar. The "Allow an app through the Windows Firewall" will appear.

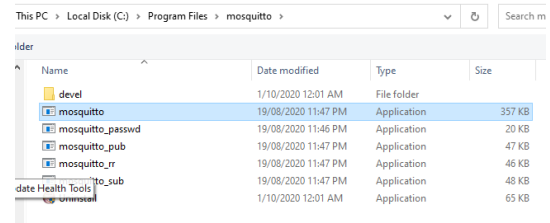


Click on this and then, on "Change Settings" and "Allow another app".

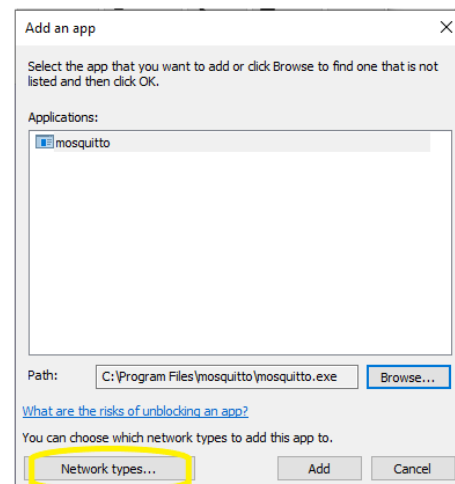


With Browse, select mosquitto.exe from C:\Program Files\mosquitto

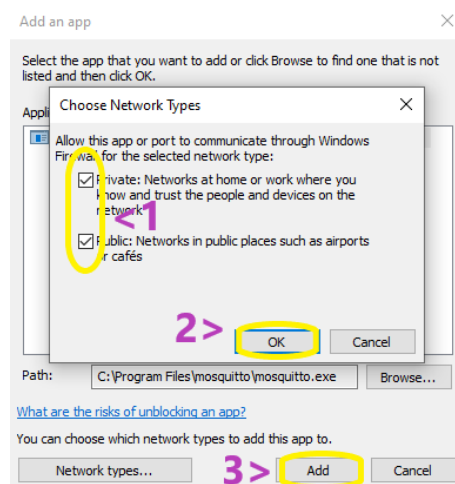
(32-bit machines would have X86 in that folder name)



And click on "Open", then click on "Network types".



Depending on how your WiFi network is configured, you may require both Private and Public options so enable both anyway. Then click on "OK" and "Add".



You will then see mosquitto added to the list. Now press OK to finish.



☒ Mixed Reality Portal

☒ mosquito

☒ MSN Weather

☒

☒

☒

Details...

Remove

Allow another app...

OK

Cancel

Credits

mqTrains uses, without modifying any of their respective code as available on github:

ArduinoJSON by Bernoit Blanchon
Arduino Queue by Einar Arnason
PubSubClient by Nick O'Leary

LittleFS file system from [earlephilhowe](#)
ESPAsyncTCP and ESPAsyncWebServer from [me-no-dev](#)